

EmoMoconvq: Emotion Conditioned Human Motion Generation

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Abstract—Physics-based motion generation methods such as MoConVQ [6] produce high-quality motions from text descriptions but generate stylistically neutral results regardless of emotional context. We propose EmoMoconvq, which extends MoConVQ’s T2M-MoConGPT by injecting a discrete emotion label (e.g., *happy*, *sad*, *angry*, *fearful*) as an additional conditioning signal. Through parameter-efficient fine-tuning on a small emotionally annotated motion dataset, we aim to generate physically plausible motions whose stylistic qualities reflect the intended emotional state.

I. MOTIVATION AND PROBLEM STATEMENT

Human motion is inherently expressive—the same action performed in different emotional states (e.g., walking while happy vs. sad) exhibits distinct kinematic patterns such as stride length, torso posture, and arm swing amplitude. This expressiveness is critical for digital humans, game characters, and social robots, yet current physics-based generation methods ignore it entirely.

MoConVQ [6] shows that discrete VQ-VAE representations combined with a GPT-style transformer can generate diverse, physically accurate motions from natural language. However, its T2M-MoConGPT produces emotion-neutral outputs: the same text prompt always yields the same stylistic result. We therefore propose **EmoMoconvq**, which extends T2M-MoConGPT by conditioning on discrete emotion labels.

II. LITERATURE REVIEW

A. Physics-Based Motion Generation.

MoConVQ [6] encodes motions into residual VQ codes and decodes them through a physics simulator via model-based RL, trained on 23 hours of diverse motion data. Its frozen encoder and codebook make it well-suited for lightweight downstream fine-tuning.

B. Text-to-Motion Generation.

T2M-GPT [7] and MDM [5] generate kinematic motions from text using discrete transformers and diffusion models respectively, both trained on HumanML3D [2]. Neither is physics-based nor models emotional style.

C. Style and Emotion in Motion.

ACTOR [4] conditions a Transformer VAE on action labels, showing that high-level categorical signals can guide motion style. The CMU MoCap database [1] includes emotionally performed locomotion sequences suitable for fine-tuning. Our work is the first to combine emotion conditioning with a physics-based generation pipeline.

III. TECHNICAL PLAN

A. System Overview

We build directly on the pretrained MoConVQ framework [6]. The MoConVQ system consists of three trained components: (1) a 1D convolutional **motion encoder** $\mathcal{E}$ that maps a motion clip M to a sequence of latent vectors $V = [v^k]$; (2) a **residual VQ codebook** $\mathcal{B}$ with $N = 8$ quantization layers, each containing 512 code vectors of dimension 768, producing discrete index sequences $S = [I^k]$; and (3) a **physics-based decoder** $\mathcal{D}$ comprising a deconvolutional network, a mixture-of-experts control policy, and a physics simulator. For text-to-motion generation, MoConVQ employs a dual-transformer **MoConGPT**: a temporal transformer ($N_T = 12$ layers) that aggregates information across motion time steps, and a depth transformer ($N_D = 5$ layers) that predicts RVQ indices layer by layer within each time step.

In our extension, we freeze all components except for a newly introduced **emotion embedding module** and the **top K layers** of the temporal transformer.

B. Emotion Embedding Module

Let $\mathcal{C} = \{\text{neutral}, \text{happy}, \text{sad}, \text{angry}, \text{fearful}\}$ denote the set of $|\mathcal{C}| = 5$ emotion categories. Each emotion $c \in \mathcal{C}$ is mapped to a trainable embedding vector:

$$E_e : \mathcal{C} \rightarrow \mathbb{R}^{d_e}, \quad e_c = E_e(c) \quad (1)$$

where $d_e = 512$ matches the hidden dimension of the temporal transformer. The emotion embedding is fused with the T5 text feature f_{T5} via a two-layer MLP projection followed by element-wise addition:

$$f_{\text{cond}} = f_{T5} + \text{MLP}_{\text{proj}}(e_c) \quad (2)$$

This additive fusion preserves the pretrained text conditioning while introducing an orthogonal affective signal. The fused feature f_{cond} replaces f_{T5} in the cross-attention layers of the temporal transformer, following the same injection mechanism as the original T2M-MoConGPT.

C. Fine-Tuning Strategy

We freeze the MoConVQ encoder, codebook, physics decoder, depth transformer, and the bottom $(N_T - K)$ layers of the temporal transformer. We fine-tune only: noitemsep

- The emotion embedding table E_e ($|\mathcal{C}| \times d_e$ parameters),
- The MLP projection head MLP_{proj} ,
- The top $K = 4$ layers of the temporal transformer (cross-attention and FFN weights).

This parameter-efficient design minimizes catastrophic forgetting of the pretrained motion distribution while allowing the model to learn emotion-specific stylistic variations. We train with the same negative log-likelihood loss as MoConGPT:

$$\mathcal{L}_{\text{emo}} = \mathbb{E}_{(S, c)} \left[\sum_{k=1}^K \sum_{d=0}^N -\log p(I_d^k \mid f_{\text{cond}}(c), I_{<d}^k, I^{<k}) \right] \quad (3)$$

D. Data Preparation

We construct a fine-tuning dataset by combining two sources:

(1) **HumanML3D subset with emotion keywords.** We filter the HumanML3D training set [2] for text annotations containing explicit emotion-related adverbs or adjectives (e.g., “happily”, “angrily”, “fearfully”, “sadly”). We assign emotion labels based on keyword matching and manual verification. We expect approximately 800–1200 clip-text pairs across the four non-neutral emotion classes.

(2) **CMU MoCap emotional gait sequences.** The CMU database includes a small set of locomotion sequences performed in specified emotional styles. We retarget these sequences to the MoConVQ character skeleton and synthesize paired text descriptions (e.g., “a person walks forward”) with assigned emotion labels. This yields approximately 100–200 additional pairs.

For data augmentation, we apply random horizontal mirroring (left/right reflection) and temporally subsample sequences at $0.8\times$ and $1.2\times$ speed, doubling the effective dataset size.

E. Evaluation Protocol

We evaluate on a held-out set of emotion-conditioned text prompts and report:

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- **FID:** Fréchet Inception Distance between generated and ground-truth motion feature distributions, computed using the pretrained motion feature extractor from [2].
- **R-Precision (Top-1/2/3):** Retrieval accuracy between generated motions and their conditioning text descriptions.
- **Emotion Classification Accuracy:** A separately trained lightweight emotion classifier (e.g., GRU-based) on motion sequences is used to verify that generated motions are classified as the intended emotion.
- **User Study:** 20 participants rate generated motion pairs (same action, different emotion labels) on a 5-point Likert scale for (a) emotional distinctiveness and (b) motion naturalness.

IV. EXPECTED OUTCOMES

We expect to demonstrate that EmoMotion successfully conditions motion style on discrete emotion labels without sacrificing the physics-based plausibility guaranteed by the MoConVQ decoder. Concretely, we anticipate: noitemsep

- FID scores competitive with or modestly higher than the baseline T2M-MoConGPT (target: ≤ 0.5 FID degradation),

- Emotion classification accuracy $\geq 60\%$ on held-out clips (significantly above the 20% random baseline for 5 classes),
- User study ratings indicating that $\geq 70\%$ of participants can correctly identify the intended emotion from the generated motion,
- Qualitative video results showing perceptibly different walking, running, and sitting motions across emotion conditions.

V. POTENTIAL RISKS AND BACKUP PLAN

A. Insufficient labeled data.

If fine-tuning overfits, we reduce to 3 emotion classes (happy/sad/neutral) and rely more heavily on augmentation.

B. Catastrophic forgetting.

If overall FID degrades sharply, we reduce K (freeze more layers) or adopt LoRA adapters [3] instead of full layer fine-tuning.

C. Compute constraints.

As a fallback, we fine-tune only the emotion embedding and MLP ($<1\text{M}$ parameters), omitting temporal transformer fine-tuning.

VI. TIMELINE

Period	Milestone
Week 1–2	Reproduce T2M-MoConGPT baseline; verify FID/R-precision.
Week 3–4	Curate emotion dataset; implement augmentation pipeline.
Week 5–6	Implement EmoMoconVq modules; run initial fine-tuning.
Week 7–8	Ablation studies; quantitative evaluation; user study.
Week 9–10	Final report and demo video.

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